PURS: Personalized Unexpected Recommender System for Improving User Satisfaction

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Main Idea of the Paper

We propose an industrial unexpected recommender system that combines four novel components:

- **Multi-Cluster Modeling of User Interests**
- **Personalized Modeling of Unexpectedness**
- **Session-based Modeling of Unexpectedness**
- **Unexpected Activation Function**

PURS model has already been deployed in Alibaba-Youku.

Unexpected Recommender System

1 item added to your bag

Samsonite Leather Slim Portfolio Laptop Briefcase

Color:
Black

$220.00
Sale $87.99
You just saved $132.01!

1 item(s) in bag **subtotal: $87.99**

you saved: $132.01

Congrats! You get **Free Shipping.**
Excludes furniture, mattresses & other exclusions apply.

View Bag & Checkout

Continue Shopping

Customers who bought this also shopped…

Samsonite Professional Lea...
Black Friday Special
Reg. $280.00
Sale $111.99

Kenneth Cole Reaction Manh...
Black Friday Special
Reg. $260.00
Sale $103.99

Ben Sherman Karino Leather...
Black Friday Special
Reg. $520.00
Sale $129.99

Kenneth Cole Reaction "Eve...
Black Friday Special
Reg. $160.00
Sale $63.99

Samsonite Classic Two Guss...
Black Friday Special
Reg. $140.00
Sale $55.99

Samsonite Leather Expandab...
Black Friday Special
Reg. $320.00
Sale $127.99

Unexpected Recommender System

Classical collaborative filtering algorithms often lead to the problem of

- Over-Specialization (Adamopoulos & Tuzhilin 2014)
- Filter Bubbles (Nguyen et al. 2014; Pariser 2011)
- User Boredom (Kapoor et al. 2015a; Kapoor et al. 2015b)

Unexpected recommender systems aim at providing novel and satisfying recommendations to the targeted users simultaneously.

Unexpected Recommender System

Three categories of methods

- Rule-based approaches: predefine rules or recommendation strategies for unexpected recommendations (Said et al. 2012)
- Model-based approaches: propose new models and data structures that go beyond the traditional collaborative filtering paradigm (Wasilewski & Hurley 2019)
- Utility-based approaches: construct the hybrid utility function as the combination of estimated relevance and degree of unexpectedness (Adamopoulos & Tuzhilin 2015)

Unexpectedness

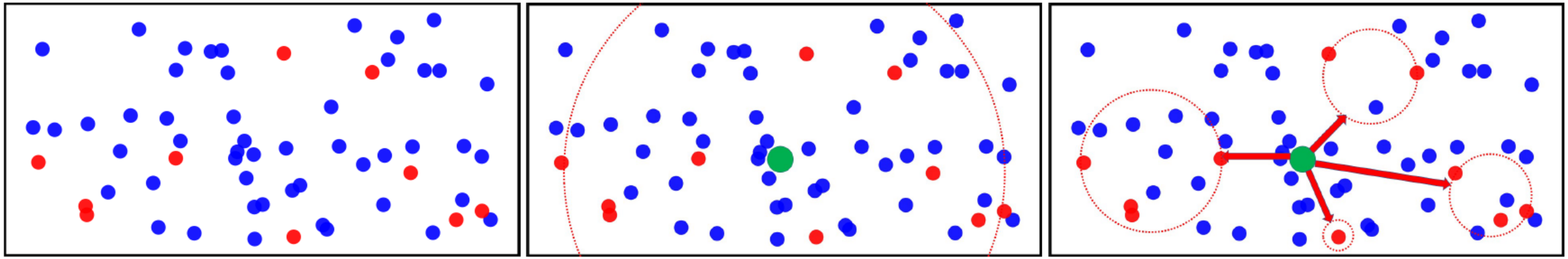
Definition. Unexpectedness of a new item as the distance between the embedding of that item and the closure of user interests in the latent space.

How to model user interests?

--Aggregation of all previously consumed item embeddings. (Li & Tuzhilin 2019)

--Clustering of all previously consumed item embeddings (This paper)

Unexpectedness



(a) Item Embeddings (b) One Single Latent Closure (c) Clustering of Latent Closure

Figure 1: Comparison of unexpectedness modeling in the latent space. Blue points stand for the available items; Orange points represent the consumed items; Green point refers to the new recommended item. We propose to define unexpectedness as the distance between the new item and clusters of latent closure generated by all previous consumptions.

Personalization & Session

Personalized Unexpectedness

- Some people tend to be "variety-seekers" and are more willing to click on novel item recommendations.
- Other people prefer to stay in their own comfort zones and are in favor of familiar item recommendations.

Session-Based Unexpectedness

- If the user has just finished the first episode, it is more reasonable to recommend the next episode of a TV series to that user, instead of recommending new types of videos to that person.
- If the user has been binge-watching the same TV series in one night, it is better to recommend to this user something different.

Utility Function

$$Utility_{u,i} = r_{u,i} + f(unexp_{u,i}) * unexp_factor_{u,i}$$

$r_{u,i}$: the click-through-rate estimation for user u towards item i based on their features and past behaviors.

$unexp_{u,i}$: the unexpectedness of item i towards user u.

$unexp_factor_{u,i}$: the personalized and session-based perception of unexpectedness for user u towards item i.

f(·): the activation function for unexpectedness in order to effectively and efficiently incorporate this piece into the utility function.

Unexpected Activation Function

The unexpected activation function f(·) should satisfy the following mathematical properties:

- **Continuous**
- **Bounded**
- **Unimodal**
- **Short-Tailed**

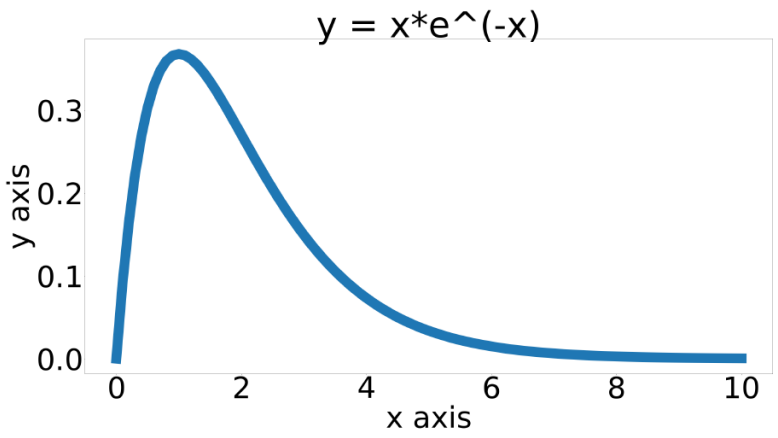
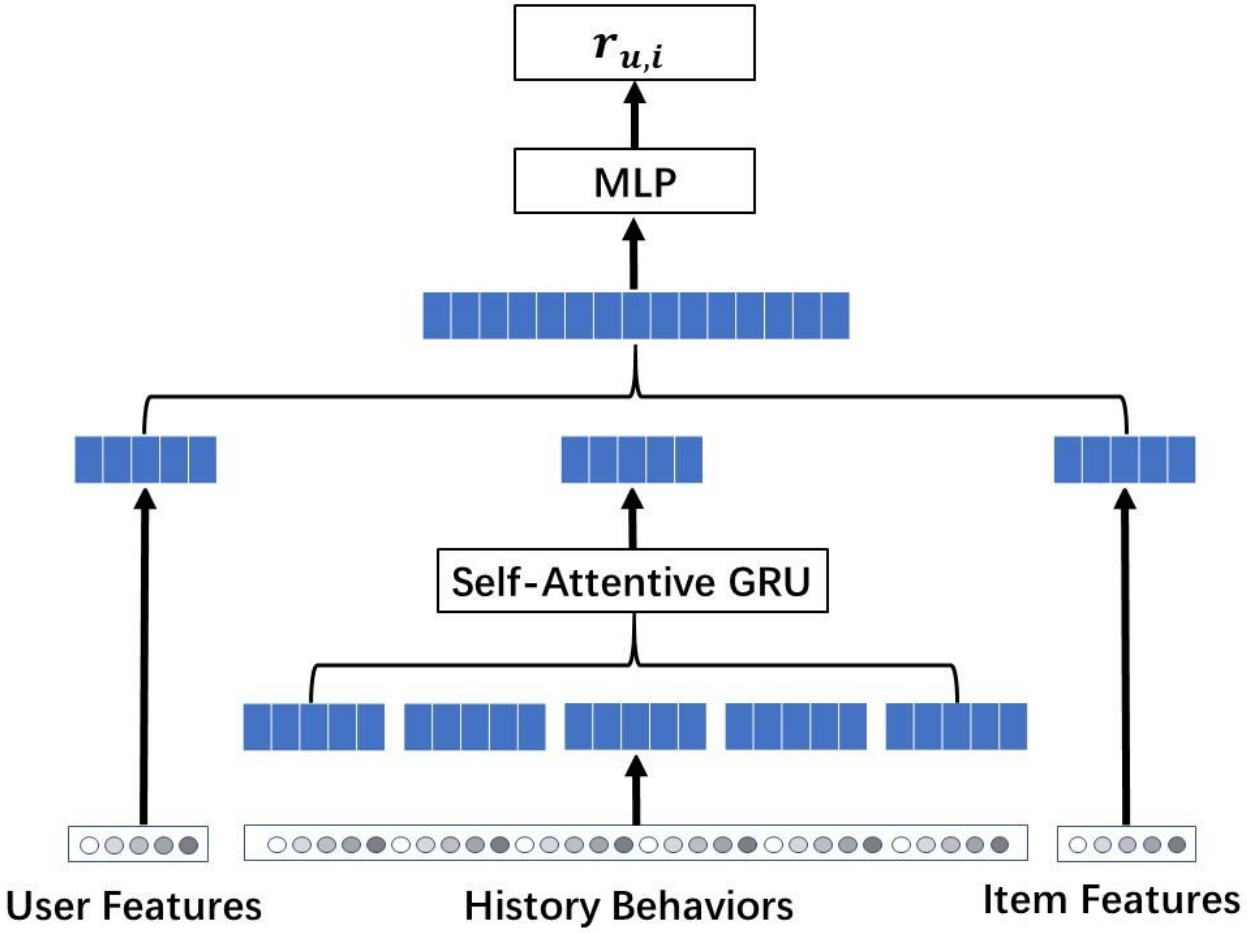
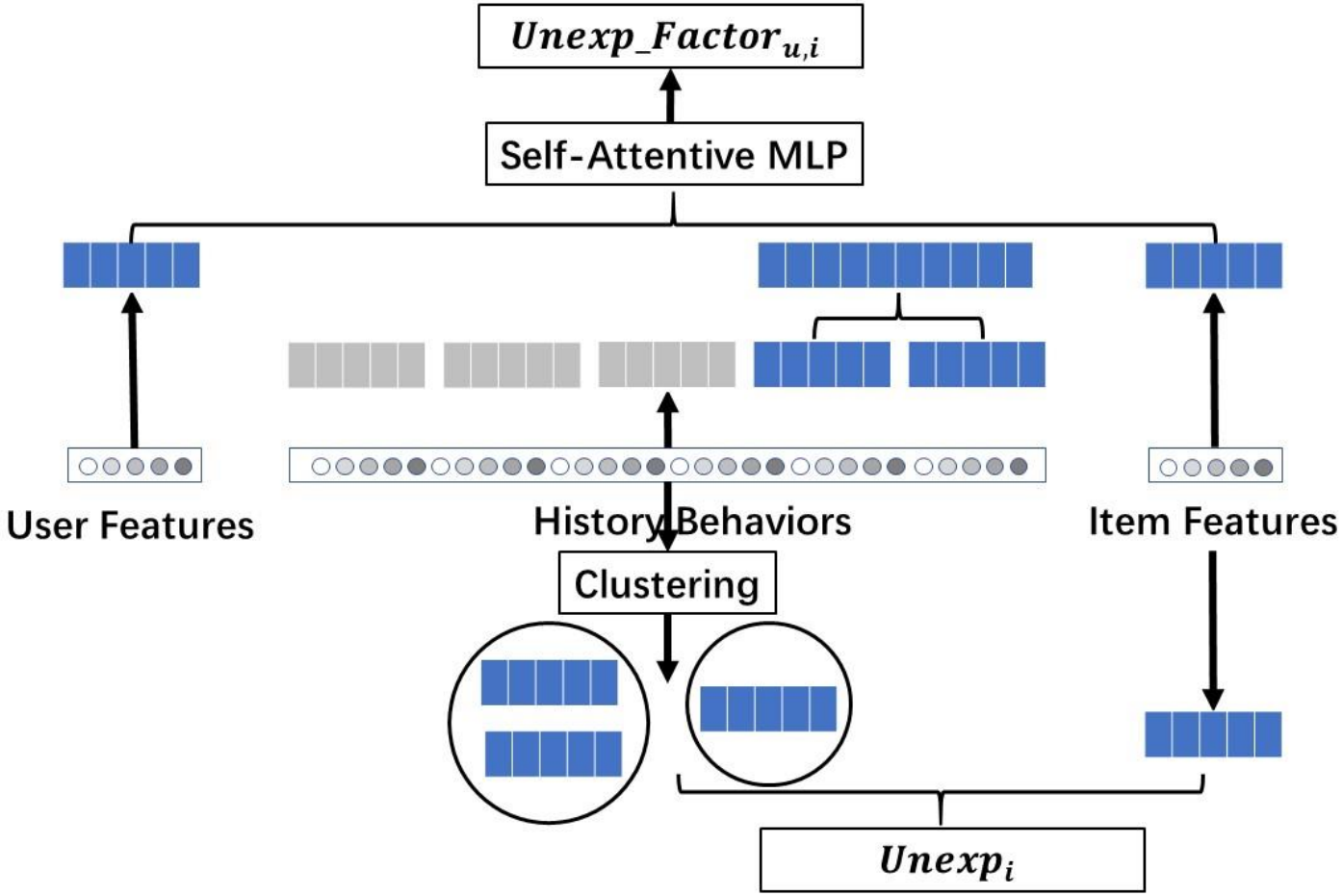


Figure 2: Unexpected Activation Function

Base Model


$r_{u,i}$
MLP
Self-Attentive GRU
User Features
History Behaviors
Item Features

Unexp Model


$Unexp_Factor_{u,i}$
Self-Attentive MLP
User Features
History Behaviors
Item Features
Clustering
$Unexp_i$

Experiments

Dataset: Yelp/MovieLens/Alibaba

- Conducted time-stratified 5-fold cross validation
- Evaluated performance through
  - Accuracy Metrics: AUC and HR@10
  - Novelty Metrics: Unexpectedness and Coverage

Dataset	Yelp	MovieLens	Alibaba
# of Records	2,254,589	19,961,113	1,806,157
# of Users	76,564	138,493	46,143
# of Items	75,231	15,079	53,657
Sparsity	0.039%	0.956%	0.073%

Experiments: Baselines

- Baseline Methods (Unexpected Recommendation):
  - **SPR**: Serendipitous Personalized Ranking (Lu et al. 2012)
  - **Auralist**: LDA-based Hybrid Recommendation (Zhang et al. 2012)
  - **DPP**: Determinant Point Process (Chen et al. 2018)
  - **HOM-LIN**: Homogeneous Users with Linear Distance (Adamopoulos & Tuzhilin 2015)
- Baseline Methods (CTR Prediction):
  - **DIN**: Deep Interest Network (Zhou et al. 2017)
  - **DeepFM**: Deep Factorization Machine (Guo et al. 2017)
  - **Wide & Deep**: Wide Network and Deep Network for Recommendation (Cheng et al. 2016)
  - **PNN:** Product-based Neural Network (Qu et al. 2008)

Results

Algorithm	Youku				Yelp				MovieLens			
	AUC	HR@10	Unexp	Coverage	AUC	HR@10	Unexp	Coverage	AUC	HR@10	Unexp	Coverage
PURS	**0.7154***	**0.7494***	**0.0913***	**0.6040***	**0.6723***	**0.6761***	**0.2401***	**0.7585**	**0.8090***	**0.6778***	**0.2719***	**0.3732***
DIN	0.6957	0.6972	0.0688	0.1298	0.6694	0.6702	0.0391	0.6934	0.7021	0.6485	0.0887	0.2435
DeepFM	0.5519	0.5164	0.0333	0.2919	0.6396	0.6682	0.0412	0.6044	0.7056	0.6169	0.1275	0.3098
Wide&Deep	0.6807	0.6293	0.0472	0.3400	0.6698	0.6693	0.0392	0.7580	0.7940	0.6333	0.0944	0.3432
PNN	0.5801	0.5667	0.0593	0.1860	0.6664	0.6692	0.0391	0.7548	0.7140	0.6382	0.1318	0.3665
HOM-LIN	0.5812	0.5493	0.0602	0.4284	0.6287	0.6490	0.1433	0.5572	0.7177	0.5894	0.1116	0.1525
Auralist	0.5319	0.5250	0.0598	0.3990	0.6428	0.6104	0.1434	0.5442	0.6988	0.5710	0.1010	0.1333
SPR	0.5816	0.5156	0.0739	0.4668	0.6364	0.6492	0.1438	0.5849	0.7059	0.6122	0.1396	0.1728
DPP	0.6827	0.5777	0.0710	0.4702	0.5940	0.6414	0.1330	0.5072	0.7062	0.5984	0.1602	0.1908

Table 2: Comparison of recommendation performance in three datasets. The first block contains baselines for click-through-rate optimization, while the second block contains baselines for unexpectedness optimization. '*' represents statistical significance at the 0.95 level.

Ablation Study

- PURS-Variation 1 (Gaussian Activation)
- PURS-Variation 2 (No Activation)
- PURS-Variation 3 (No Unexpectedness Factor)
- PURS-Variation 4 (No Unexpectedness)
- PURS-Variation 5 (Single Closure of User Interest)

Algorithm	Youku				Yelp				MovieLens			
	AUC	HR@10	Unexp	Coverage	AUC	HR@10	Unexp	Coverage	AUC	HR@10	Unexp	Coverage
PURS	**0.7154***	**0.7494***	**0.0913***	**0.6040***	**0.6723***	**0.6761***	**0.2401***	**0.7585***	**0.8090***	**0.6778***	**0.2719***	**0.3732***
PURS-Variation 1	0.6826	0.7292	0.0828	0.5548	0.6682	0.6602	0.1298	0.7292	0.7757	0.6419	0.1812	0.3350
PURS-Variation 2	0.7067	0.7148	0.0707	0.3026	0.6692	0.6630	0.0412	0.7580	0.8041	0.6585	0.2471	0.3580
PURS-Variation 3	0.7036	0.6720	0.0762	0.1522	0.6508	0.6692	0.0391	0.7583	0.7666	0.6460	0.0888	0.2792
PURS-Variation 4	0.7063	0.5628	0.0688	0.1298	0.6702	0.6702	0.0391	0.6934	0.7586	0.6331	0.0887	0.2435
PURS-Variation 5	0.7038	0.7080	0.0477	0.2042	0.6701	0.6700	0.0395	0.7580	0.7715	0.6596	0.1727	0.3561

Table 3: Ablation Study of recommendation performance in three datasets. '*' represents statistical significance at the 0.95 level.

Online A/B Test

- **Test Period: November 2019 to December 2019**

Business Metrics:

- **VV**: Video View, average video viewed by each user
- **TS**: Time Spent, average time that each user spends on the platform
- **ID**: Impression Depth, average impression through one session
- **CTR**: Click-Through-Rate

	VV	TS	ID	CTR	Unexpectedness	Coverage
Improvement	+3.74%*	+4.63%*	+4.13%*	+0.80%*	+9.74%*	+1.23%*

Table 4: Unexpected recommendation performance in online A/B test: performance increase compared to the current model. '*' represents statistical significance at the 0.95 level.

Conclusions

- **We propose an industrial unexpected recommender system PURS that is capable of providing novel and satisfying recommendations simultaneously.**
- **We conduct extensive offline and online experiments to demonstrate the power of PURS, leading to its industrial deployment.**

Q&A

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